

Diffusive Regime for the High-Frequency Dynamics of Randomly Heterogeneous Plates

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Summary. We use radiative transfer equations to describe the high-frequency vibrations of randomly heterogeneous plates. The diffusive regime is depicted both numerically and analytically from this model. Numerical simulations are based on the Monte-Carlo method, whereas the analytical model is obtained by an asymptotic analysis. The objectives are to improve the understanding of high-frequency vibrations of complex industrial structures.

1 Introduction

Realistic, complex industrial structures exhibit typical transport and diffusive behaviors in the high-frequency range of vibration [1]. Whereas their dynamics are physically well represented by global standing waves (the eigenmodes) in the low-frequency range, this approach fails dramatically as frequency increases, because of the multiple interactions of waves with material heterogeneities at the wavelength scale, and reflections on the boundaries and at the interfaces between structural components with different stiffnesses (stiffeners on plates and shells, bulkheads, stringers ...). Predictive methods for high-frequency vibrations are needed in the automotive and aerospace industries for applications to noise control, acoustic comfort and discretion, or pyrotechnic shock simulation for instance. Statistical Energy Analysis (SEA, see [2]) is the most popular method among mechanical engineers to deal with such problems. It is based on some rigorous arguments developed in the late 50s [3], but also on several phenomenological assumptions which can hardly be satisfied exactly. Derivation of a heat conduction analogy for high-frequency structural dynamics has attracted several researchers in the early 90s [4], but this model is based on heuristic, unproven arguments as well.

In [5], we have shown that the radiative transfer model proposed in [6] can be used to analyze the high-frequency vibrations of randomly heterogeneous plates by an energetic approach. Expressions for the scattering cross-sections of shear and bending motions have also been derived. This model is based on the assumptions that (i) wavelengths are short with respect to the macroscopic features of the medium (the high-frequency range), (ii) they are comparable to the correlation lengths of random heterogeneities, and (iii) fluctuations of the latter are small and rather isotropic. In this paper we

focus on the diffusive regime obtained both numerically and analytically. Indeed diffusion equations can be derived from transport equations in a rather systematic way for long propagation distances with respect to the scattering mean free path (see Ryzhik *et al.* [6] and references therein). In the next section we recall our previous results on transport for Mindlin plates and show some numerical simulations exhibiting a diffusive behavior. The radiative transfer equations are solved numerically by the Monte-Carlo method [7]. Diffusion is described analytically in Section 3.

2 Radiative transfer equations

Theoretical background

An isotropic Mindlin plate occupies the plane $\{z = 0\}$ ($\mathbf{x} = (x, y)$) and we denote by ϱ its mass density, E its Young's modulus, G its shear modulus such that $G = E/2(1 + \nu)$, ν its Poisson's coefficient and $\kappa < 1$ its shear reduction coefficient. We consider here the case of an homogeneous constant-velocity background medium with random heterogeneities on the scale of the wavelength. Fluctuations of the material properties at this scale cause a wave with wave vector $\mathbf{k}'$ at a point $\mathbf{x}$ to be scattered into any direction $\hat{\mathbf{k}}$ with wave vector $\mathbf{k}$. The multigroup radiative transfer equation for shear and bending energies is [6, 5]

$$\mathbf{s} \partial_t \mathbf{a}(\mathbf{x}, \mathbf{k}, t) + \hat{\mathbf{k}} \cdot \nabla_{\mathbf{x}} \mathbf{a}(\mathbf{x}, \mathbf{k}, t) = \int_{R^2} \boldsymbol{\sigma}(\mathbf{k}, \mathbf{k}') \mathbf{a}(\mathbf{x}, \mathbf{k}', t) d\mathbf{k}' - \boldsymbol{\Sigma}(\mathbf{k}) \mathbf{a}(\mathbf{x}, \mathbf{k}, t) \quad (1)$$

where $\mathbf{s}$ is the slowness matrix, $\boldsymbol{\sigma}$ is the kernel of the collision operator and $\boldsymbol{\Sigma}$ is the total scattering matrix:

$$\mathbf{s} = \text{diag} \left(\frac{1}{c_T}, \frac{1}{c_P}, \frac{1}{c_S} \right), \quad \boldsymbol{\sigma} = \begin{bmatrix} \sigma_{TT} & 0 & 0 \\ 0 & \sigma_{PP} & \sigma_{PS} \\ 0 & \sigma_{SP} & \sigma_{SS} \end{bmatrix}, \quad \boldsymbol{\Sigma} = \text{diag}(\Sigma_T, \Sigma_P, \Sigma_S). \quad (2)$$

$\mathbf{a} = (a_T, a_P, a_S)$ is the vector of the angularly resolved energy densities, or specific intensities, for shear – one mode labelled T – and bending – two single modes labelled P and S – motions. The wave speeds for shear and bending are respectively $c_T^2 = \kappa c_S^2$ and $c_P^2 = (\frac{2}{1-\nu})c_S^2$ with $c_S^2 = \frac{G}{\varrho}$. Random fluctuations of parameters ϱ , $1/\kappa G$ and $1/E$ are modeled by second order, homogeneous stochastic processes described by their power spectral density functions $\hat{R}_\varrho(\mathbf{k})$, $\hat{R}_{\kappa G}(\mathbf{k})$ and $\hat{R}_E(\mathbf{k})$ respectively, and cross-spectral density functions $\hat{R}_{\varrho/\kappa G}(\mathbf{k})$ and $\hat{R}_{\varrho/E}(\mathbf{k})$. Scattering cross-sections are then given by (see [5] for details on the derivation):

$$\sigma_{TT}(\mathbf{k}, \mathbf{k}') = \frac{\pi}{2} c_T |\mathbf{k}|^2 \left[\hat{R}_\varrho(\mathbf{k} - \mathbf{k}') + (\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')^2 \hat{R}_{\kappa G}(\mathbf{k} - \mathbf{k}') \right. \\ \left. + 2(\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}') \Re \hat{R}_{\varrho/\kappa G}(\mathbf{k} - \mathbf{k}') \right] \delta(c_T |\mathbf{k}| - c_T |\mathbf{k}'|), \quad (3)$$

$$\sigma_{PP}(\mathbf{k}, \mathbf{k}') = \frac{\pi}{2} c_P |\mathbf{k}|^2 \left[(\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')^2 \hat{R}_\varrho(\mathbf{k} - \mathbf{k}') + [(1 - \nu)(\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')^2 + \nu]^2 \hat{R}_E(\mathbf{k} - \mathbf{k}') \right. \\ \left. + 2(\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}') [(1 - \nu)(\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')^2 + \nu] \Re \hat{R}_{\varrho/E}(\mathbf{k} - \mathbf{k}') \right] \delta(c_P |\mathbf{k}| - c_P |\mathbf{k}'|), \quad (4)$$

$$\sigma_{SS}(\mathbf{k}, \mathbf{k}') = \frac{\pi}{2} c_S |\mathbf{k}|^2 \left[(\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')^2 \hat{R}_\varrho(\mathbf{k} - \mathbf{k}') + [2(\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')^2 - 1]^2 \hat{R}_E(\mathbf{k} - \mathbf{k}') \right. \\ \left. + 2(\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}') [2(\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')^2 - 1] \Re \hat{R}_{\varrho/E}(\mathbf{k} - \mathbf{k}') \right] \delta(c_S |\mathbf{k}| - c_S |\mathbf{k}'|), \quad (5)$$

$$\sigma_{PS}(\mathbf{k}, \mathbf{k}') = \frac{\pi}{2} c_P \left(1 - (\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')^2 \right) \left[(1 - \nu)^2 |\mathbf{k}'|^2 (\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')^2 \hat{R}_E(\mathbf{k} - \mathbf{k}') \right. \\ \left. + |\mathbf{k}|^2 \hat{R}_\varrho(\mathbf{k} - \mathbf{k}') + 2(1 - \nu) |\mathbf{k}| |\mathbf{k}'| (\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}') \Re \hat{R}_{\varrho/E}(\mathbf{k} - \mathbf{k}') \right] \delta(c_P |\mathbf{k}| - c_S |\mathbf{k}'|) \quad (6)$$

and $\sigma_{SP}(\mathbf{k}, \mathbf{k}') = \sigma_{PS}(\mathbf{k}', \mathbf{k})$, with $\mathbf{k} = |\mathbf{k}| \hat{\mathbf{k}}$ and $\mathbf{k}' = |\mathbf{k}'| \hat{\mathbf{k}}'$. We have assumed that ν remains deterministic but this assumption is not restrictive. Finally the total scattering cross-sections are given by:

$$\Sigma_\alpha(\mathbf{k}) = \sum_\beta \int_{R^2} \sigma_{\alpha\beta}(\mathbf{k}, \mathbf{k}') d\mathbf{k}', \quad \alpha, \beta \in \{T, P, S\}. \quad (7)$$

Numerical Simulations by the Monte-Carlo Method

The radiative transfer equation (1) is solved numerically by the Monte-Carlo method [7]. We simulate particle trajectories in the homogeneous background medium such that the rays are its geodesic lines with (i) a Poisson distribution for the estimation of the time of the next scattering on random heterogeneities, and (ii) the distribution of outgoing velocities and polarization at this time given by the (normalized) scattering cross-sections. Thus the probability distribution for a particle to jump from wave vector $\mathbf{k}$ and mode α to wave vector $\mathbf{k}'$ and mode β is given by

$$P(\mathbf{k}, \alpha; d\mathbf{k}', \beta) = \frac{\sigma_{\alpha\beta}(\mathbf{k}, \mathbf{k}') d\mathbf{k}'}{\Sigma_\alpha(\mathbf{k})}. \quad (8)$$

Boundary conditions are taken into account in our simulations with the assumption that the rays are reflected according to Snell-Descartes' law (lossless reflections).

Figure 1 displays our results for bending energetic waves with the following data: the spectral density functions $\hat{R}_{jk}(\mathbf{k})$ are given by

$$\hat{R}_{jk}(\mathbf{k}) = \frac{h_j h_k}{4\pi\gamma^2} e^{-\frac{|\mathbf{k}|^2}{4\gamma^2}}$$

with $h_\varrho = 0.1$ and $h_E = 1$ for fluctuation amplitudes, Poisson's coefficient is $\nu = 0.2$ and $c_S = 1$. For such a correlation model, scattering cross-sections are

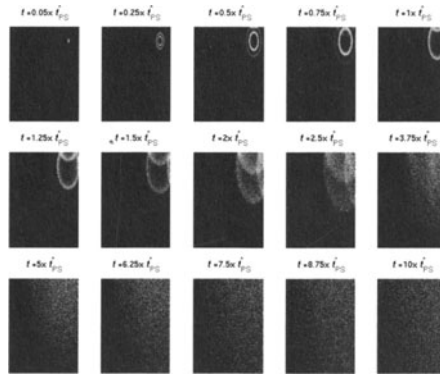


Fig. 1. Bending energy waves for Gaussian correlations of parameters ($k_0 = 10$)

rotationally invariant, that is they depend on $|\mathbf{k}|$ and $\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}'$ solely. The dimensionless wave number is denoted by $k_0 = \gamma^{-1}|\mathbf{k}|$ where γ^{-1} is the correlation length of parameters. The size of the plate is 4×6 the maximum diffusion mean free path, $\max(\ell_P^*, \ell_S^*)$, defined in the next section. Time steps are multiples of the diffusion time $t_{PS}^* = \max(\ell_P^*/c_P, \ell_S^*/c_S)$. The source is a unit point load applied at time $t = 0$ in the upper right corner. We observe that a diffusive regime is attained at higher frequencies ($k_0 = 10$) with a rather uniform energy spread in the plate. This regime is described analytically in the next section.

3 The Diffusive Regime

Derivation of radiative transfer equations has been carried out so far for high frequencies such that $\Lambda \ll c_\alpha t$ and they remain applicable as long as $\Lambda \ll \ell_\alpha = \frac{c_\alpha}{\Sigma_\alpha}$, where Λ is a typical wavelength and ℓ_α is the scattering mean free path for the mode α . The diffusion limit considers that propagation distances are furthermore much longer than the latter: $c_\alpha t \gg \ell_\alpha$. Transported waves exhibit a diffusive behavior inasmuch as their direction of propagation is lost after travelling several scattering mean free paths. Also they become approximately depolarized in this regime. This situation corresponds to the conditions in which SEA is applied for bounded structures. In this case, the scattering mean free path is small with respect to the characteristic dimensions of the structure. From such a diffusion equation, we can obtain proportionality relations between the energy densities of particular propagative modes and their associated velocities, similarly to the ones given between vibrational energies and modal densities in SEA.

The diffusive regime is characterized by a diffusion mean free path ℓ_α^* , which is the distance waves must travel until their direction is randomized. For single-mode propagation, such as shear waves in a Mindlin plate, the

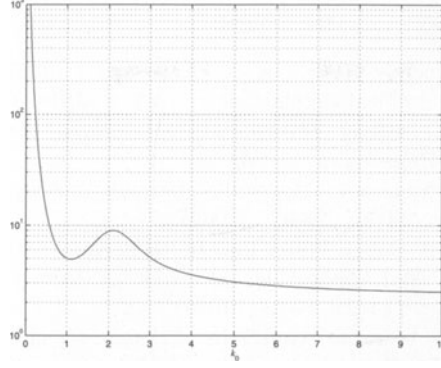


Fig. 2. Normalized diffusion coefficient $D_{\text{PS}}^{\#}$ for Gaussian correlations of parameters

specific intensity a_{T} is independent of $\hat{\mathbf{k}}$ and satisfies the diffusion equation

$$\frac{\partial a_{\text{T}}(\mathbf{x}, |\mathbf{k}|, t)}{\partial t} = \nabla_{\mathbf{x}} \cdot [D_{\text{T}}(|\mathbf{k}|) \nabla_{\mathbf{x}} a_{\text{T}}(\mathbf{x}, |\mathbf{k}|, t)] \quad (9)$$

where $D_{\text{T}} = \frac{1}{2} c_{\text{T}} \ell_{\text{T}}^*$ is the diffusion coefficient. For multi-modal propagation, such as bending waves in a Mindlin plate, specific intensities a_{P} and a_{S} are linked by an equipartition rule $a_{\text{P}}(\mathbf{x}, |\mathbf{k}|, t) = a_{\text{S}}(\mathbf{x}, \frac{c_{\text{P}}}{c_{\text{S}}} |\mathbf{k}|, t) := \Psi(\mathbf{x}, |\mathbf{k}|, t)$ such that

$$\frac{\partial \Psi(\mathbf{x}, |\mathbf{k}|, t)}{\partial t} = \nabla_{\mathbf{x}} \cdot [D_{\text{PS}}(|\mathbf{k}|) \nabla_{\mathbf{x}} \Psi(\mathbf{x}, |\mathbf{k}|, t)] . \quad (10)$$

The diffusion coefficient is now

$$D_{\text{PS}}(|\mathbf{k}|) = \frac{c_{\text{P}} c_{\text{S}}}{2(c_{\text{P}}^2 + c_{\text{S}}^2)} \left(c_{\text{S}} \ell_{\text{P}}^*(|\mathbf{k}|) + c_{\text{P}} \ell_{\text{S}}^* \left(\frac{c_{\text{P}}}{c_{\text{S}}} |\mathbf{k}| \right) \right) \quad (11)$$

where the diffusion mean free paths are solutions of

$$\begin{bmatrix} \Sigma_{\text{P}}(|\mathbf{k}|) - \Sigma_{\text{PP}}^*(|\mathbf{k}|) & -\Sigma_{\text{PS}}^*(|\mathbf{k}|) \\ -\Sigma_{\text{SP}}^* \left(\frac{c_{\text{P}}}{c_{\text{S}}} |\mathbf{k}| \right) & \Sigma_{\text{S}} \left(\frac{c_{\text{P}}}{c_{\text{S}}} |\mathbf{k}| \right) - \Sigma_{\text{SS}}^* \left(\frac{c_{\text{P}}}{c_{\text{S}}} |\mathbf{k}| \right) \end{bmatrix} \begin{pmatrix} \ell_{\text{P}}^*(|\mathbf{k}|) \\ \ell_{\text{S}}^* \left(\frac{c_{\text{P}}}{c_{\text{S}}} |\mathbf{k}| \right) \end{pmatrix} = \begin{pmatrix} c_{\text{P}} \\ c_{\text{S}} \end{pmatrix}$$

with $\Sigma_{\alpha\beta}^*(|\mathbf{k}|) = \int_{S^1} \tilde{\sigma}_{\alpha\beta}(|\mathbf{k}|, \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}') \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}' d\hat{\mathbf{k}}'$ (S^1 is the unit sphere of R^2). Here and for the scalar shear mode we have assumed that scattering cross-sections are rotationally invariant: $\sigma_{\alpha\beta}(\mathbf{k}, \mathbf{k}') = \frac{c_{\beta}}{|\mathbf{k}'|} \tilde{\sigma}_{\alpha\beta}(|\mathbf{k}|, \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}') \delta(c_{\alpha} |\mathbf{k}| - c_{\beta} |\mathbf{k}'|)$. Figure 2 displays the normalized diffusion coefficient $D_{\text{PS}}^{\#} = \gamma D_{\text{PS}} / c_{\text{S}}$ for the same data as in the previous section.

We note to conclude that the results obtained thus far can account for in-plane motions as well and can be extended to the case of Timoshenko beams, cylindrical shells or porous layers modeled by Biot's equations. Research is in progress to include a viscous damping model with memory effects in the current analysis.

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